EE2001 Electrical Circuits- I:

Continuation to:

Simple Resistive Circuits

17 October 2023, Tuesday

Outline

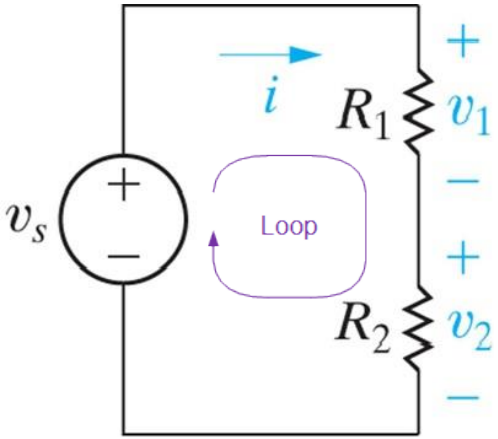


3. The Voltage-Divider and Current-Divider Circuits

4. Voltage Division and Current Division

The Voltage-Divider Circuit

- Especially in **electronic circuits**, usually **two or more voltage levels** need to be developed from **a single voltage supply**.
- **One way of** doing it is **voltage-divider circuit**.



$$v_s = v_1 + v_2 \qquad v_s = R_1 i_s + R_2 i_s$$

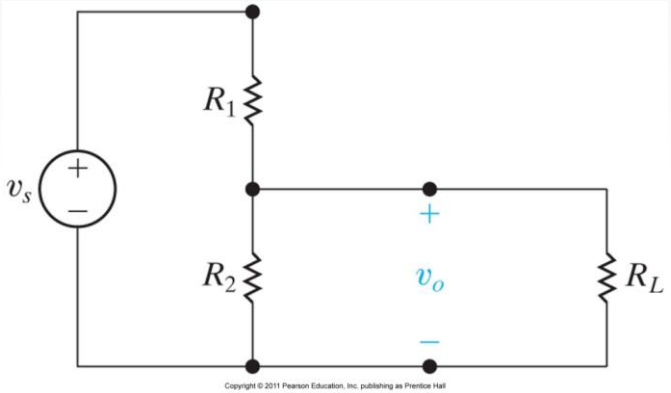
$$i_s = \frac{v_s}{R_1 + R_2}$$

$$v_1 = R_1 i_s = \frac{R_1}{R_1 + R_2} v_s$$

$$v_2 = R_2 i_s = \frac{R_2}{R_1 + R_2} v_s$$

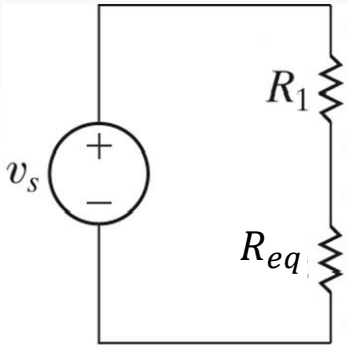
The Voltage-Divider Circuit

- Sometimes **external circuits** can be connected to **a divided voltage level**:



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$\Rightarrow$



$$v_o = \frac{R_{eq}}{R_1 + R_{eq}} v_s$$

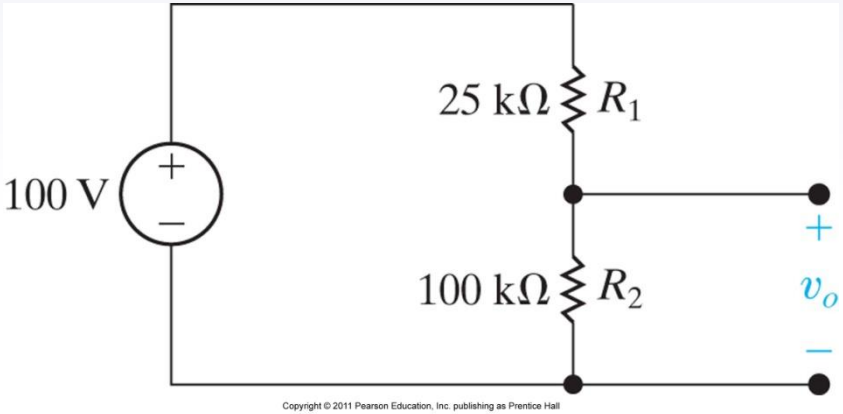
$$R_{eq} = \frac{R_2 R_L}{R_2 + R_L}$$

$$v_o = \frac{R_2}{R_1(1 + R_2 / R_L) + R_2} v_s$$

- As $\boldsymbol{R_L \to \infty}$, $\boldsymbol{V_o}$ **becomes** $\boldsymbol{V_2}$ (since less current will flow over R_L).
- As long as $\boldsymbol{R_L \gg R_2}$, $\boldsymbol{v_o/v_s}$ **is disturbed very little** by the addition of the load on the divider.
- The **tolerances** of **resistors in the market** can also **affect** the output voltage of a divider.

The Voltage-Divider Circuit

- An example



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If the resistors used in the voltage-divider have a tolerance of ±10%, find the maximum and minimum value of $\boldsymbol{v_o}$.

Maximum value occurs when $\boldsymbol{R_1}$ is 10% low (22.5kΩ) and $\boldsymbol{R_2}$ is 10% high (110 kΩ):

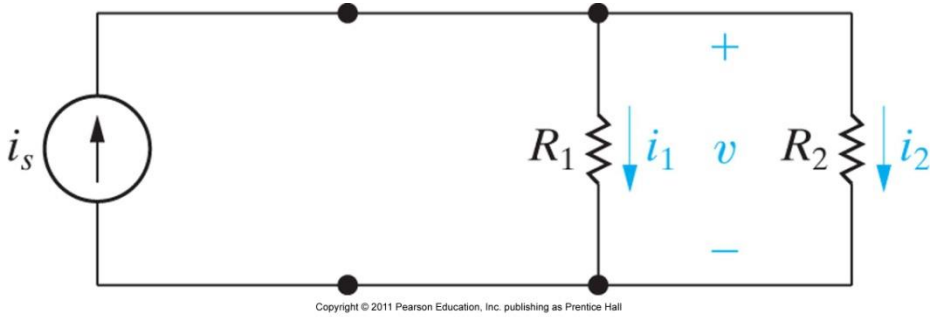
$$v_{o(max)} = \frac{(100)(110)}{110 + 22.5} = 83.02\ V$$

Minimum value occurs when $\boldsymbol{R_1}$ is 10% high (27.5kΩ) and $\boldsymbol{R_2}$ is 10% low (90 kΩ):

$$v_{o(min)} = \frac{(100)(90)}{90 + 27.5} = 76.60\ V$$

3 The Current-Divider Circuit

- In a current-divider circuit with two parallel resistors, the current flowing through each resistors is inversely proportional to the values of the resistors respectively.



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$$i_s = i_1 + i_2 \qquad i_s = \frac{v}{R_1} + \frac{v}{R_2}$$

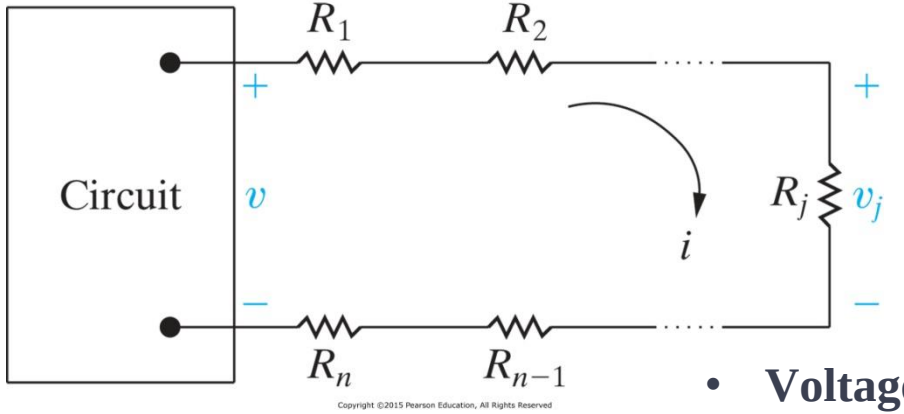
$$v = \frac{R_1 R_2}{R_1 + R_2} i_s$$

$$i_1 = \frac{v}{R_1} = \frac{R_2}{R_1 + R_2} i_s$$

$$i_2 = \frac{v}{R_2} = \frac{R_1}{R_1 + R_2} i_s$$

4 Voltage and Current Division

- We can **generalize the formulas** for voltage-divider and current-divider circuits with larger number of resistors.
- For a **voltage-divider** circuit with ***n* resistors**:



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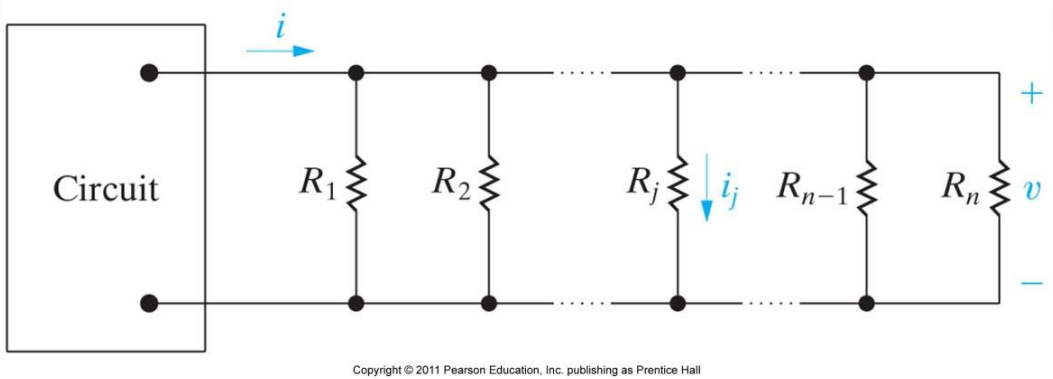
$$i = \frac{v}{R_1 + R_2 + R_3 + \ldots + R_n} = \frac{v}{R_{eq}}$$

$$v_j = R_j i = \frac{R_j}{R_{eq}} v = \frac{R_j}{R_1 + R_2 + R_3 + \ldots + R_n} v$$

- **Voltage on a resistor** is **proportional** to the **ratio** of **its resistance** value to the **equivalent resistance.**

4 Voltage and Current Division

- For a **current-divider** circuit with ***n* resistors**:



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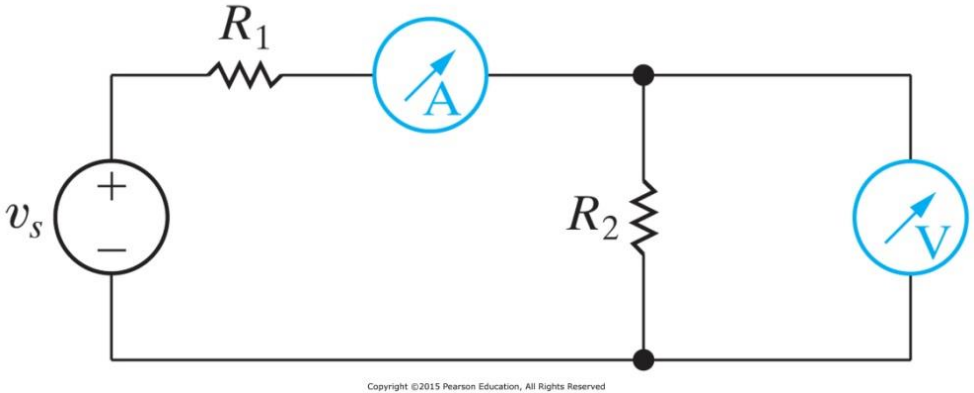
$$v = R_{eq} i$$

$$i_j = \frac{v}{R_j} = \frac{R_{eq}}{R_j} i$$

- **Current flowing through a resistor** is **proportional** to the **ratio** of the **equivalent resistance** to **its resistance.**

5 Measuring Voltage and Current

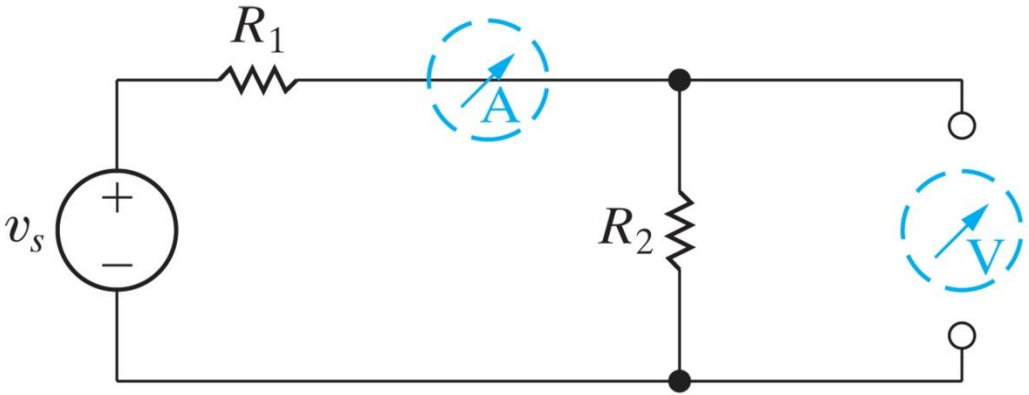
- A **voltmeter** is used to measure voltages. It is connected **in parallel** to the branch **across** which the voltage will be measured.
- An **ampermeter** is used to measure current. It is connected **in series** to the branch **through** which the current will be measured.



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5 Measuring Voltage and Current

- An **ideal** measuring device **does not have an effect** on the currents and voltages in the circuit.
- An **ideal voltmeter** has **infinite** resistance (can be considered as an **open circuit** case).
- An **ideal ampermeter** has **zero** resistance (can be considered as a **short circuit** case).



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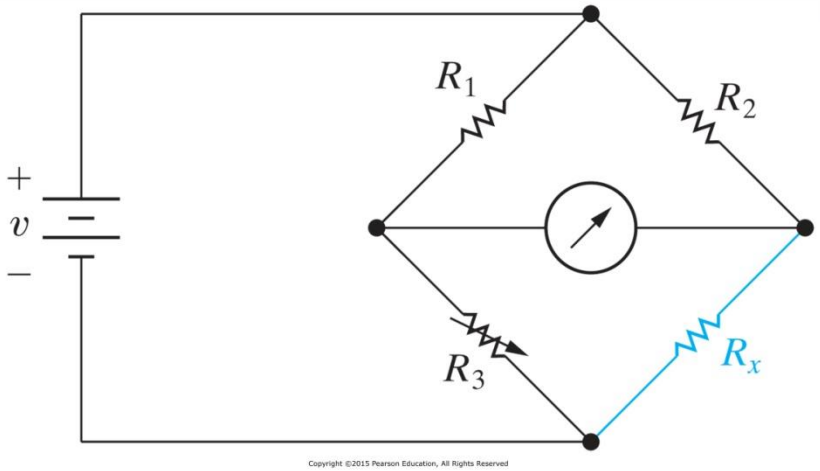
5 Measuring Voltage and Current

- Voltmeters and ampermeters can be **digital** or **analog**.
- Actual voltmerers and ampermeters are **not ideal**. Therefore, they can have **some effect** on the circuit (i.e., they can **disturb** the **currents** and **voltages** in the circuit).
- In order to have **reliable measurements**, the **internal resistance** of a **voltmeter** must be **much larger than** the other resistances in the circuit.
- In contrast, in order to have **reliable measurements**, the **internal resistance** of an **ampermeter** must be **much smaller than** the other resistances in the circuit.

6

Measuring Resistance: The Wheatstone Bridge

- The Wheatstone Bridge is a circuit used to measure **resistances of medium values** (from **1Ω to 1MΩ**).

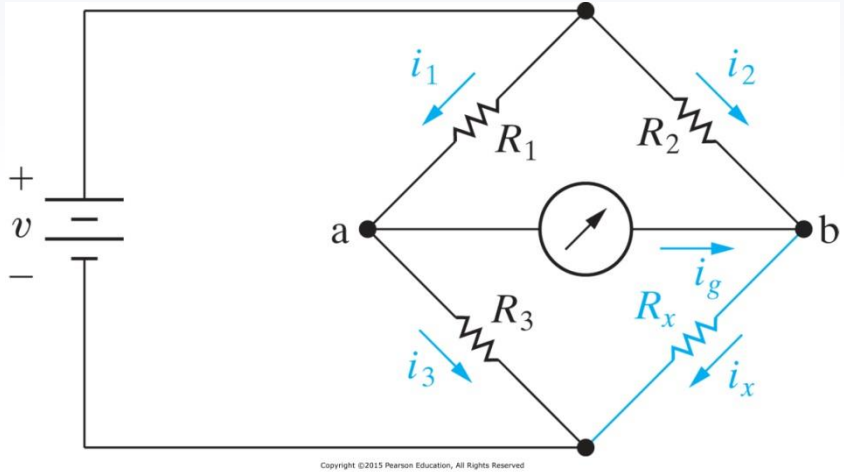


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- R_1, R_2 and R_3 are **known,** R_x is **unknown**.
- R_3 is a **variable resistor** (its value can be **adjusted**).
- The **galvanometer** is a d'Arsonval meter in the **microamper (μA) range**.
- Usually **a battery** is used as a source.

Measuring Resistance: The Wheatstone Bridge

- **Vary** the R_3 resistor **until there is no current** in the **galvanometer**.



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$$i_g = 0 \qquad v_{ab} = 0$$

$$i_1 = i_3 \qquad i_2 = i_x$$

$$R_1 i_1 = R_2 i_2 \qquad R_3 i_3 = R_x i_x$$

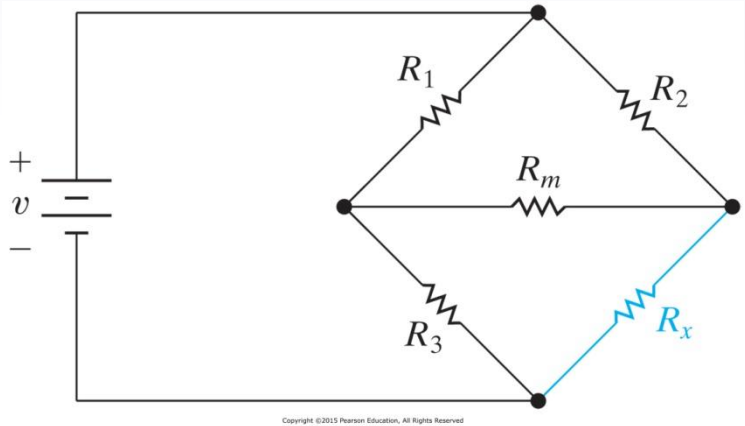
$$R_3 i_1 = R_x i_2 \qquad \frac{R_1}{R_3} = \frac{R_2}{R_x}$$

$$R_x = \frac{R_2}{R_1} R_3$$

- If $R_1 = R_2$ then $\boldsymbol{R_x = R_3}$.
- By varying R_1, R_2 and R_3, **different ranges** for measuring R_x can be obtained.

7 Delta-to-Why (Pi-to-Tee) Equivalent Circuits (Δ-to-Y (π-to-T) Equivalent Circuits)

- Consider the Wheatstone bridge circuit with the **galvanometer replaced by a resistance.**

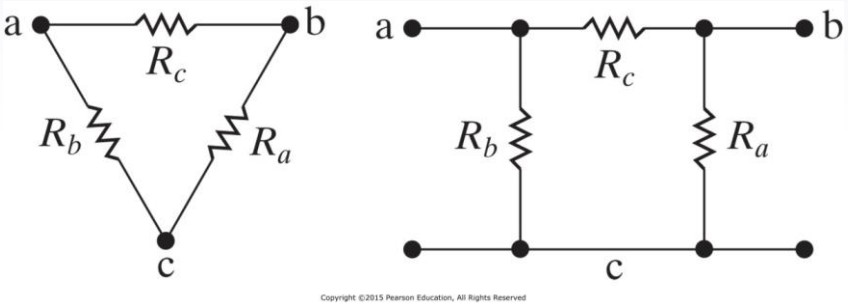


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- Some of the resistors here are **neither just series nor just parallel**.
- By using **only the rules of parallel or series** resistances, the circuit **cannot be reduced** to a single equivalent resistor.

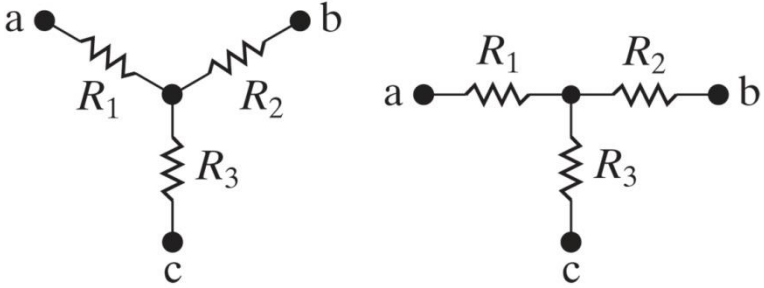
7 Delta-to-Why (Pi-to-Tee) Equivalent Circuits (Δ-to-Y (π-to-T) Equivalent Circuits)

- The interconnection given below is called **delta (Δ)** interconnection or **pi (π)** interconnection:



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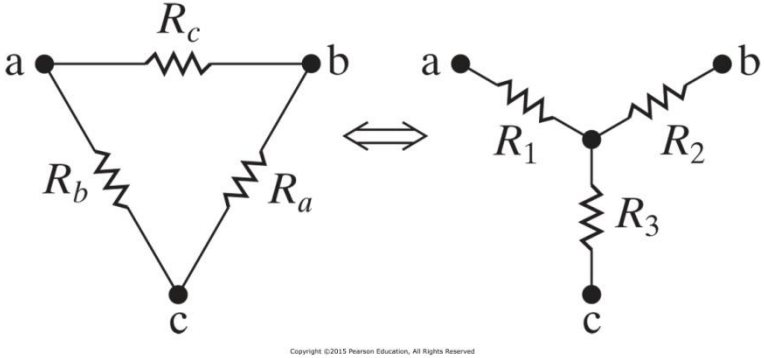
- The interconnection given below is called **why (Y)** interconnection or **tee (T)** interconnection:



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7 Delta-to-Why (Pi-to-Tee) Equivalent Circuits (Δ-to-Y (π-to-T) Equivalent Circuits)

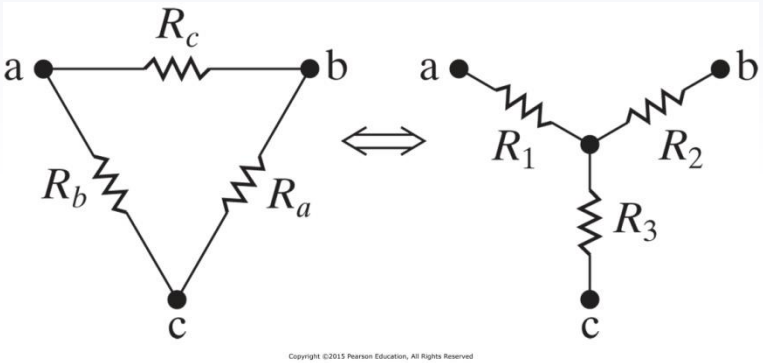
- For the analysis of some circuits sometimes it is better to **replace** one of these interconnection types with its **equivalent of the other type**.
- Such replacement may significantly **simplify the calculations**, so it is useful to derive their equivalents.



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- By equivalent, it is meant that the **terminal behavior** (the current and voltages at the terminals) of the two circuits are the same. For this reason, **equivalent resistance between each pair of terminals** of the two circuits must be the same.

Delta-to-Why (Pi-to-Tee) Equivalent Circuits (Δ-to-Y (π-to-T) Equivalent Circuits)



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$$R_{ab} = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} = R_1 + R_2$$

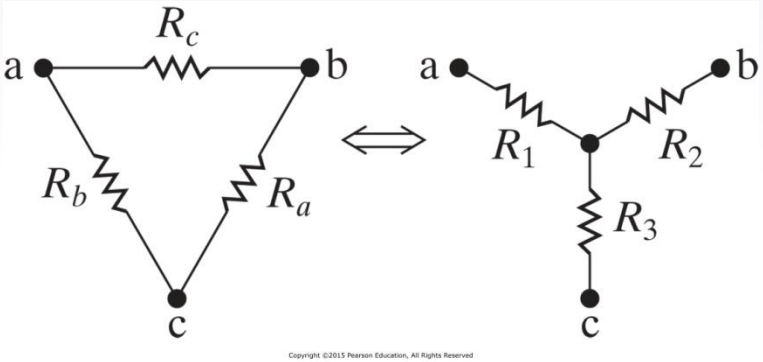
$$R_{bc} = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} = R_2 + R_3$$

$$R_{ca} = \frac{R_b(R_c + R_a)}{R_a + R_b + R_c} = R_1 + R_3$$

- Its is possible to solve the equations for R_1, R_2 and R_3 **in terms of R_a, R_b and R_c.**
- It is also possible to solve the equations for R_a, R_b and R_c **in terms of R_1, R_2 and R_3.**

Delta-to-Why (Pi-to-Tee) Equivalent Circuits (Δ-to-Y (π-to-T) Equivalent Circuits)

- Solving for $\boldsymbol{R_1, R_2}$ **and** $\boldsymbol{R_3}$:



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$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

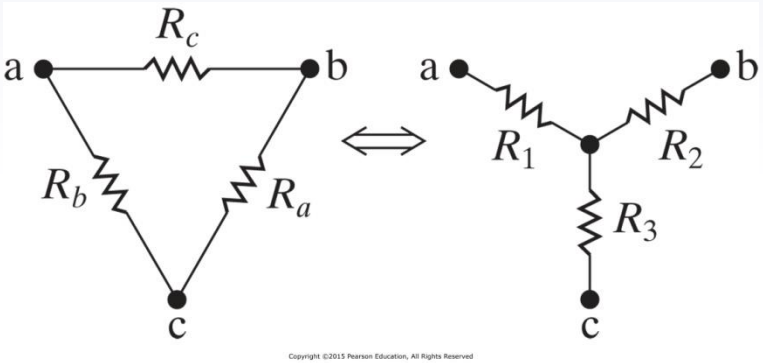
$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

- These are the values of the resistors in the Y-connection in terms of the values of the resistors in the Δ-connection.

Delta-to-Why (Pi-to-Tee) Equivalent Circuits (Δ-to-Y (π-to-T) Equivalent Circuits)

- Solving for R_a, R_b and R_c:



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$$R_1R_2 + R_2R_3 + R_1R_3 = \frac{R_aR_bR_c}{R_a + R_b + R_c}$$

$$R_a = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_1}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_2}$$

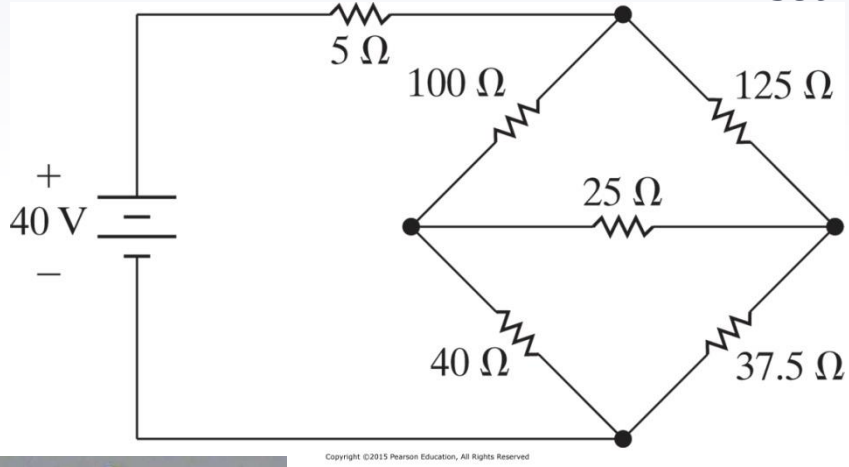
$$R_c = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_3}$$

- These are the values of the resistors in the Δ-connection in terms of the values of the resistors in the Y-connection.

Delta-to-Why (Pi-to-Tee) Equivalent Circuits (Δ-to-Y (π-to-T) Equivalent Circuits)

- Example

Find the **current** and **power** supplied by the 40 V source in the circuit.

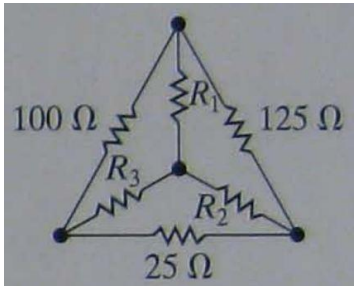


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The equivalent resistance **has to be known** to calculate the current and power.

It can be **easily found** after replacing the upper Δ or the lower Δ with its equivalent Y.

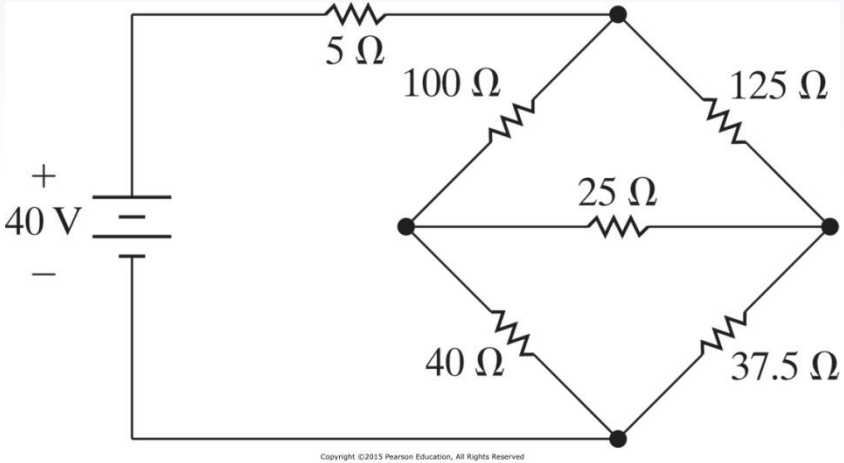
We choose to **replace the upper Δ**:



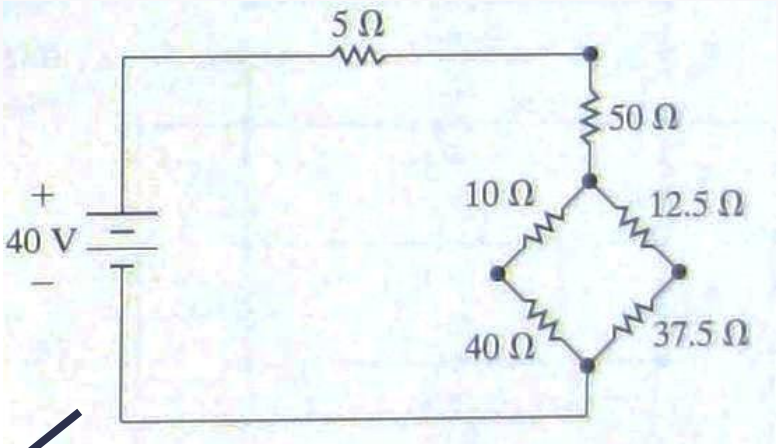
$$R_1 = \frac{100x125}{250} = 50\ \Omega \quad R_2 = \frac{125x25}{250} = 12.5\ \Omega \quad R_3 = \frac{100x25}{250} = 10\ \Omega$$

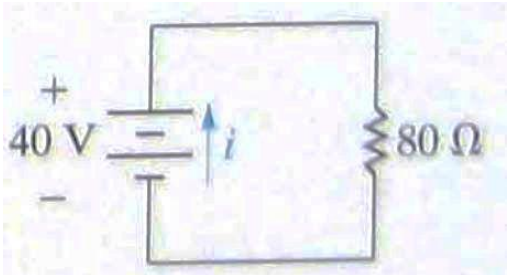
Delta-to-Why (Pi-to-Tee) Equivalent Circuits (Δ-to-Y (π-to-T) Equivalent Circuits)

- Example



Substituting the Y-resistors into the circuit:





$$R_{eq} = 5 + 50 + \frac{50x50}{100} = 80\ \Omega$$

$$i = \frac{40}{80} = 0.5\ A$$

$$P = (0.5)40 = 20\ W$$

End of the Lesson (17.10.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist

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